# Paleomagnetism of the Niobrara Formation in Wyoming, Colorado, and Kansas

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None of the approximately 350 samples collected from the Upper Cretaceous Niobrara formation in Wyoming, Colorado, and Kansas are reversely magnetized. Of these, 109 are completely oriented samples from limestone and chalk units; the remainder are shale samples oriented only with respect to the vertical. Stability is indicated by the determination of reliable Cretaceous pole positions (near 65°N, 170°W) from completely oriented limestone samples and by the satisfactory performance of some shale samples in a fold test. Micropaleontological studies show that the Niobrara formation ranges in age from the Coniacian to the early Maestrichtian (about 88–69 m.y. B. P.), and so this study provides further support for the existence of long periods of normal polarity of the geomagnetic field during the Cretaceous. This result is difficult to reconcile with the geomagnetic polarity time scale extrapolated from linear marine magnetic anomalies, which indicates several reversals during this period of time.

During the last 10 yr or so, many studies have been made of the geomagnetic polarity history of the earth's magnetic field. Information from these studies can be a valuable tool in such diverse areas as sea floor spreading [*Vine and Matthews*, 1963], stratigraphic correlation [*Burek*, 1970], and the mechanism of the earth's magnetic field [*Cox*, 1969]. There are presently several techniques for extending this polarity history back in time as far as possible, each with certain advantages and disadvantages. Studies of radiometrically dated igneous rocks are useful when the time interval under consideration is well represented by such rocks and when the error in radiometric age determination is less than the length of the shortest polarity interval requiring resolution [*Cox and Dalrymple*, 1967]. Extrapolation of the geomagnetic polarity history from linear magnetic anomaly patterns [*Heirtzler et al.*, 1968] often proceeds upon the assumption that sea floor spreading is continuous, an idea that it is desirable to test independently. A third technique is the study of sedimentary rocks. Paleomagnetic studies of older sedimentary rocks have concentrated on red beds [*Runcorn*, 1955; *Gose and Helsley*, 1972] because they are generally intensely magnetized and fairly stable after demagnetization. Disadvantages of red bed study are that (1) many time intervals worth studying are not represented by good red bed sections and (2) many red bed sections are unfossiliferous or contain fossils that are such poor temporal indicators that the sections can only be dated on the basis of stratigraphic relationships. Because of these limitations it is desirable to extend such studies to nonred sediments, which form the bulk of the sedimentary record and which (particularly for the case of limestones and shales) often contain abundant microfossils allowing more precise dating.

Intensive paleomagnetic studies of limestone and shale sections are rare mainly because intensities of magnetization are generally so low ($10^{-6}$ emu/cm$^3$ or lower) that they can only be measured with considerable difficulty [*Strangway and McMahon*, 1973]. Lack of study has left the question of whether or not these rocks provide stable carriers of primary remanence largely unanswered. These points are some of the problems to which we address ourselves in this study of the Niobrara formation.

The Niobrara formation is an Upper Cretaceous sequence of limestones and shales that occurs throughout much of the western interior of the United States. The samples for this study come from three different localities (see index map, Figure 1): Centennial Valley east of Laramie, Wyoming (41.25°N, 106.21°W), three locations in Pueblo, Colorado (38.27°N, 104.68°W; 38.26°N, 104.65°W; 38.31°N, 104.66°W), and Trego County, Kansas (38.78°N, 100.06°W). In addition, for purposes of comparison 13 samples were collected near Kemmerer, Wyoming (41.85°N, 110.51°W) from the Hilliard formation, a shale and sandstone sequence that is at least partly the temporal equivalent of the Niobrara formation in western Wyoming.

The Niobrara sections show considerable variation in the three localities sampled in this study (Figure 2). At Pueblo the entire section is about 230 m thick [*Scott and Cobban*, 1964]. The most heavily sampled unit is the basal Fort Hays limestone member, which is about 12 m thick and consists of limestone beds averaging 0.3 m in thickness separated by thin layers of calcareous shale. The rest of the units are predominantly shale with thinner units of limestone and chalk. At the Trego County section in Kansas the Fort Hays limestone was the only unit sampled. Here it is about 20 m thick and more massive than that at Pueblo, with individual limestone beds up to 1.5 m thick [*Hattin*, 1965]. There is no equivalent to the Fort Hays limestone at the Centennial Valley section (in fact, no limestone units exist there at all). The 210-m-thick section is dominated by the Sage Breaks shale member and the upper shale member; carbonates are represented in the lower and upper chalk members [*Atherton*, 1971]. The Hilliard formation at the Kemmerer site is about 900 m thick and consists mainly of shale [*Adams*, 1972]. Most of the paleomagnetic samples from this location come from three sandstone units near the middle of the section.

In conjunction with the paleomagnetic study reported in this paper, detailed micropaleontological studies have been made on the sections at all four localities. This work [*Frerichs and Adams*, 1973; *Adams and Frerichs*, 1973] has made it possible to date the sections quite well with respect to each other and with respect to the Upper Cretaceous section of Texas following the zonation of *Pessagno* [1967]. This information is summarized in Figure 2, which indicates that our samples span a period of about 20 m.y. from Coniacian to early Maestrichtian.

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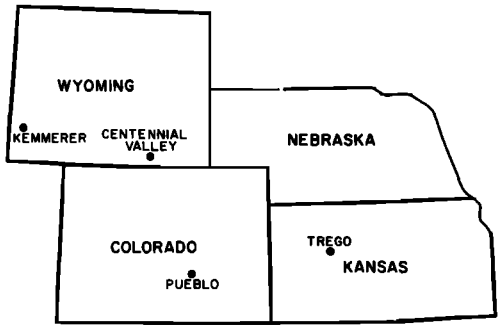



Fig. 1. Index map showing the location of the four localities described in this study.

## PROCEDURE

Similar collection techniques were used at all four locations. When the rock was not prone to splitting or parting, hand samples oriented by Brunton compass were collected at the outcrop. Standard paleomagnetic cores were drilled from these samples in the laboratory; in most cases at least two specimens were obtained from each hand sample. Most limestones, sandstones, and chalks were collected by this method. Shaley units could not be drilled in the field or the laboratory unless special impregnation procedures were used before drilling. These rock types were collected by cutting bedding plane slabs into disks that were stacked into cylindrical plastic containers 3.2 cm in diameter by 2.5 cm high. The orientation of the disks with respect to top and bottom was preserved, but the azimuth was not. These samples could only be used for determinations of normal or reversed polarity, whereas the completely oriented samples yielded information about polarities as well as virtual geomagnetic poles.

The sampling interval is different for the four sections studied as well as for different rock types within a section. We consider that the Pueblo and Centennial Valley sections are the most important parts of our study. At these locations the sampling interval was about 0.3 m in limestone and chalk units and about 1.0–1.5 m in shale units. The sections at Kemmerer and Trego, which were used mainly for purposes of comparison, were not sampled as completely. Figure 2 shows the distribution of completely and partly oriented samples at each location. Poor exposures of some units at Pueblo are responsible for some lack of coverage there.

All measurements of remanence were made on a Schonstedt SSM-1 spinner magnetometer. This instrument has a theoretical sensitivity of $4 \times 10^{-9}$ emu/cm$^3$ (for a 10-cm$^3$ sample), but practical considerations such as sample holder magnetization and the time required to measure a weakly magnetized sample restricted most of our work to samples with intensities of at least $5 \times 10^{-8}$ emu/cm$^3$. A four-axis tumbler and field control system were used for alternating field demagnetization of the completely oriented cores. The samples in the cylindrical plastic boxes were demagnetized only along the cylindrical axis because they were too large to tumble and because magnetization along this axis was the only information desired from these samples. The earth's field was normal to the cylinder axis during demagnetization of these samples, and they were demagnetized in both directions (cylinder axis rotated 180°) to guard against and check for the acquisition of anhysteretic remanence.

## RESULTS

The most significant finding of this study is that none of the 323 samples are reversely magnetized. That is, none are clearly magnetized in a direction more than 90° different from the paleofield for that site with north pole at 65°N, 175°W, a

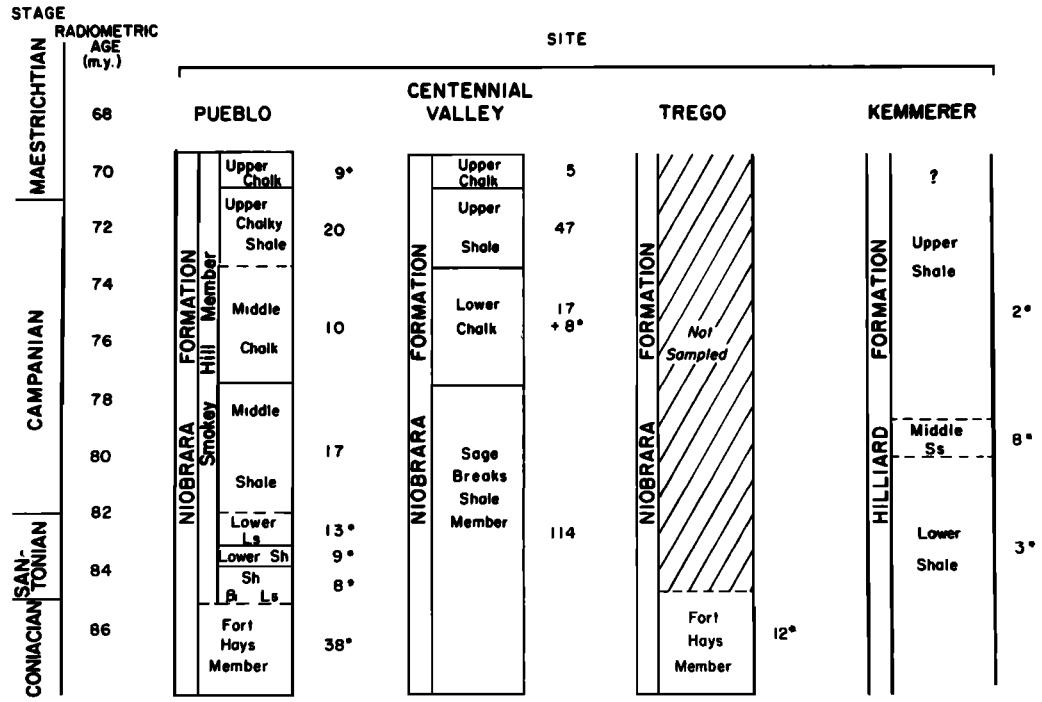



Fig. 2. Stratigraphic sections, ages, and sample coverage at the four sites. Age determinations were made by micropaleontological studies of samples from all localities (see text for references). Dashed boundaries are approximate. Relative lengths of Cretaceous stages and corresponding radiometric ages are taken from *Gill and Cobban* [1966]. Numbers to the right of the columns refer to the number of samples from the corresponding units. Asterisks refer to completely oriented samples.

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reasonable average Cretaceous pole for North America [*Grommé et al.*, 1967]. We will return to a discussion of the validity and significance of this result in a later section, but first we will discuss the other magnetic results in more detail.

*Centennial Valley.* More samples (196) were collected at this locality than at any of the other three. All but eight were collected in plastic boxes, since even the chalk members (Figure 2) were so shaley that collection of block samples was difficult. All shale samples were demagnetized to at least 100 Oe. As was stated before, none of the specimens were reversely magnetized. The oriented samples were different from those collected at Pueblo and Trego in that they were chalks (rather than limestones), were more poorly exposed and more weathered, and had such low natural remanent magnetization (NRM) intensities that demagnetization was only carried out to 50 Oe. The mean direction of magnetization after 50-Oe demagnetization of the eight oriented samples ($D = 296°$, $I = 76°$) corresponds to a virtual geomagnetic pole (VGP) at 48°N, 144°W. The small number of samples and large dispersion ($k = 4.5$, $\alpha_{95} = 23°$) indicate that little weight should be given to this result, but the direction of magnetization is much closer to the Cretaceous paleofield direction after tectonic correction than to the present-day field direction before tectonic correction.

*Pueblo.* A total of 124 samples (77 completely oriented) were collected at this locality. Most oriented samples consist of two specimens. A group of 30 samples was chosen for statistical analysis. All but six of these are from the Fort Hays member and from the shale and limestone unit of the Smoky Hill member. Selection of a sample depended upon its satisfying either of two criteria: (1) that the NRM intensity be greater than $2 \times 10^{-6}$ emu/cm$^3$ and (2) that the intensity after demagnetization at 200 Oe be greater than $1 \times 10^{-7}$ emu/cm$^3$. These criteria were designed to isolate samples that were more stably and more intensely magnetized than the average. Measurements on such samples are less subject to errors due to instrument noise and sample holder magnetization. The analyses are based on the magnetization present after 100-Oe demagnetization. Paleomagnetic results are shown in Table 1 together with results from completely oriented samples from other sites. The pole position for this group (66°N, 173°W) is very close to the previous Cretaceous estimate, and the $\alpha_{95}$ of 5.9° is the smallest for any of the four localities (although this situation is largely due to the greater number of samples used in the analysis). None of the limestone or shale samples are reversely magnetized.

Twenty-two of the 46 samples from the Fort Hays limestone member and the overlying shale and limestone unit of the Smoky Hill member failed the above selection criteria. Several of these had NRM intensities less than $1 \times 10^{-8}$ emu/cm$^3$ and were not demagnetized. The VGP for these 22 samples is at 79°N, 131°W, and $k$ is 7.2. The remaining 25 oriented samples that were not used in the statistical analysis were collected from the lower shale, lower limestone, and upper chalk units (Figure 2). The Niobrara formation at Pueblo becomes progressively more chalky toward the top, and these units resemble in lithology and magnetic behavior the Centennial Valley chalks more than the more massive limestones of the lower units at Pueblo. The average NRM intensity for these 25 samples is $0.57 \times 10^{-7}$ emu/cm$^3$, the pole for this group plots at 70°N, 121°W, and $k$ is 4.1.

If all samples had consisted of three or four specimens, we would have used a selection criterion based on within-sample dispersion (as was done for the Trego samples). An intensity-stability criterion is perhaps the next best, but we feel that any reasonably discriminating criterion will favor limestone samples over chalk samples. It is clear that the chalks that we have studied are less suitable candidates for paleomagnetic pole determinations. We suspect that their greater porosity and permeability allow deeper and faster weathering. For example, it is not unusual to find a yellow or orange staining of limestone and chalk outcrops alike, but the limestones are usually stained only on the surface and along joints, whereas the chalks are often stained throughout.

*Trego.* At this locality, 12 completely oriented samples were collected from the Fort Hays limestone member. Each sample consists of three or four specimens. Samples were selected for paleomagnetic analysis on the basis of the within-sample dispersion criterion that the $\alpha_{95}$ for the several specimens from a sample be less than 15° after demagnetization at 100 Oe. Five samples satisfied this criterion, and their statistics are shown in Table 1. The pole position agrees quite well with that for the Pueblo locality, and the precision ($k = 90$) is the highest for any site.

The NRM intensities of all Trego samples fall between $1 \times 10^{-7}$ to $4 \times 10^{-7}$ emu/cm$^3$, being much more uniform but slightly less intense than the Pueblo limestone samples used for statistical analysis. If we had selected Trego samples on the basis of intensity, only one different sample would have been included in the group of five, the result being a shift of 2° and an increase in the $\alpha_{95}$ of 1°. The seven samples excluded from the analysis had within-sample $\alpha_{95}$ values ranging from 18.6° to 72°, a VGP at 80°N, 132°W, and a $k$ of 7.7.

*Kemmerer.* Thirteen completely oriented samples were collected from the Hilliard formation at this site. Eight of these were from the middle sandstone unit; the remaining five are from sandy beds in the upper and lower shale units. Only one specimen was measured per sample, and those with intensities greater than $6 \times 10^{-7}$ emu/cm$^3$ after demagnetization at 100 Oe (11 in all) were chosen for analysis. Table 1 shows that the VGP for this site is in good agreement with pole positions for the Pueblo and Trego sites, although the smaller number of samples and specimens per sample has increased the dispersion.

TABLE 1. Mean Directions and Virtual Geomagnetic Poles of Samples From the Four Localities of This Study

| Locality | Mean Direction | | | | | Virtual Pole | | | |
|---|---|---|---|---|---|---|---|---|---|
| | $N$ | $k$ | $\alpha_{95}$, deg | $D$, deg | $I$, deg | Longitude, °W | Latitude, °N | $\delta_p$, deg | $\delta_m$, deg |
| Pueblo | 30 | 18.5 | 5.9 | 329 | 62 | 173 | 66 | 7.6 | 9.2 |
| Centennial Valley | 8 | 4.5 | 23.0 | 296 | 76 | 142 | 42 | 39.0 | 42.0 |
| Trego | 5 | 90.0 | 6.6 | 327 | 62 | 166 | 65 | 8.1 | 10.2 |
| Kemmerer | 11 | 10.9 | 12.8 | 320 | 64 | 182 | 62 | 16.3 | 20.3 |

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*Stability tests.* Alternating field demagnetization was carried out on almost all samples and was the only laboratory test for stability. Demagnetization characteristics of an 'average' limestone or an 'average' shale are difficult to assess from these tests, however, because there is sampling bias in the field and because the less intense samples are often removed from the population as demagnetization reduces the intensities to values too small to yield useful data. All the Trego samples plus 23 samples from Pueblo (distributed fairly uniformly along the limestone units) were demagnetized at 100 Oe, and a histogram of the intensities before and after demagnetization is shown in Figure 3*a*. Similarly, 38 shale samples from Centennial Valley and Pueblo were demagnetized at 100 and 200 Oe (Figure 3*b*). In this figure, intensities less than $1/16 \times 10^{-7}$ emu/cm$^3$ are not accurately known and may in fact be less than $1/32 \times 10^{-7}$ emu/cm.$^3$ The effects of sample bias may be seen in Figure 3*c*, which shows histograms for the Pueblo samples that were still above $1 \times 10^{-7}$ emu/cm$^3$ after demagnetization at 200 Oe. The resistance to demagnetization at these low fields of limestones and shales is similar, but the former are more than twice as strongly magnetized. By comparison, the chalks from Centennial Valley have an average NRM of about $0.7 \times 10^{-6}$ emu/cm$^3$, even less than the average for shales. The sandstone samples have an average NRM of about $1 \times 10^{-6}$ emu/cm$^3$.

The effect of demagnetization on the direction of magnetization is surprisingly small. Analysis of the NRM of all completely oriented samples from Pueblo yields a VGP at 74°N, 144°W, which suggests some viscous magnetization in the present earth's field, but analysis of the NRM of only those 30 samples that survived the selection criteria shows a VGP at 68°N, 171°W and an $\alpha_{95}$ of 6.4°, virtually identical to the result after 100-Oe demagnetization shown in Table 1. The same is true of the five samples selected from Trego, whose NRM values indicate a pole only 4° away from that determined for the demagnetized samples and a $k$ of 34, which is not significantly different from the value of 90 for the demagnetized samples, using the maximum $F$ ratio test [*Irving*, 1964]. Apparently reliable paleomagnetic results can be obtained from the NRM measurements of many samples; unfortunately, these samples cannot be identified before demagnetization.

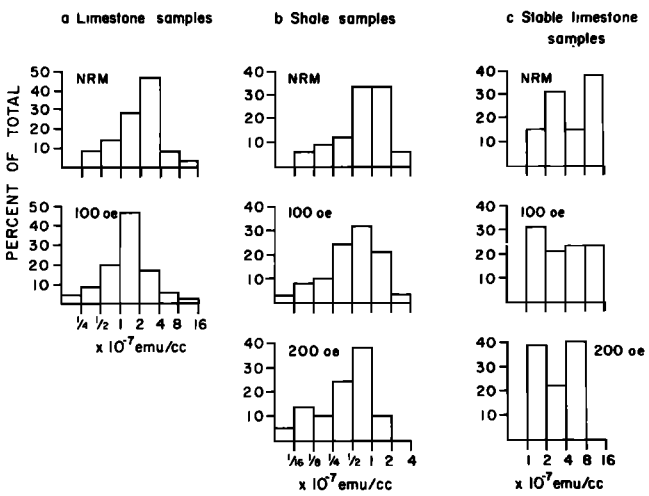



Fig. 3. Alternating field demagnetization spectra of different rock types: (*a*) limestone samples from Trego and Pueblo, (*b*) shale samples from Centennial Valley and Pueblo, and (*c*) limestone samples from Pueblo with intensities greater than $10^{-7}$ emu/cm$^3$ after demagnetization at 200 Oe. Note change of scale of intensity axis for all three groups.

*Tilt corrections.* The evidence for stability provided by the determination of Cretaceous rather than present-day pole positions is strengthened when the attitudes of the outcrops are taken into account. The Trego section is essentially horizontal, and the Pueblo section dips only about 2°–3° to the east, but the Centennial Valley and Kemmerer sections are both steeply dipping. At Centennial Valley the chalk outcrops dip about 42° toward S60°W, whereas the sandstones at the Kemmerer section dip about 47° to the west. The Centennial Valley pole, uncorrected for tilt, is at 57°N, 19°W, whereas the uncorrected pole for the Kemmerer site is at 48°N, 24°W. Both corrected poles are much closer to the Cretaceous average (4° and 20° angular separation for Kemmerer and Centennial Valley, respectively) than the uncorrected poles are to the present-day pole (38° and 31° angular separation, respectively), this situation being an indication of at least partial stability.

*Fold test for shales.* Since the shale samples were not oriented with respect to declination, stability criteria based on direction of magnetization could not be used. The shales have resistance to demagnetization that is similar to that of the limestones (Figure 3), but this is itself not a good indicator of stability. In order to test the stability better we collected 16 completely oriented shale samples from the Centennial Valley area and applied a fold test. Centennial Valley is a syncline; the east limb strikes N30°W and dips 42°W, and the west limb strikes N-S and is overturned slightly, dipping about 85°W. The above strikes and dips are averages, since local tight folding may cause sudden changes in attitude. Eight block samples from each limb were oriented (with local strike and dip noted) and brought back to the laboratory, where they were soaked in a polyester resin under vacuum and drilled into standard cores. Two or three cores were obtained from each sample except for two samples from each limb that were destroyed at various stages of the preparation and drilling process. All specimens from the 12 remaining samples were demagnetized at 50 and 100 Oe.

Figure 4 shows the directions of magnetization of the 12 samples after 100-Oe demagnetization before and after fold correction. The uncorrected directions clearly form separate populations quite different from the present-day field direction and quite obviously controlled by the local direction of folding. The fold correction causes the two populations to merge quite well, and we conclude that the magnetization has been stable in these samples at least since they were folded in the Paleocene. We also found that demagnetization has little effect on the remanent directions; Figure 4 would look almost identical if the NRM directions had been used. Unfortunately, it is dangerous to infer from this test a similar stability for all the shale samples in this study. Samples for this test had more tendency to remain in one piece than most shales that we collected. This behavior may mean that they were generally less weathered and thus less subject to the decay of the original remanence and the overprinting of secondary components.

A puzzling feature of this test is that the corrected directions do not cluster about a reasonable direction for the Cretaceous paleofield. Considerable errors were certainly introduced during sample orientation in the field, measurement of local dip and strike, and specimen orientation after drilling, but these are more likely to be random than systematic. Other possibilities include (1) that a truly anomalous field direction

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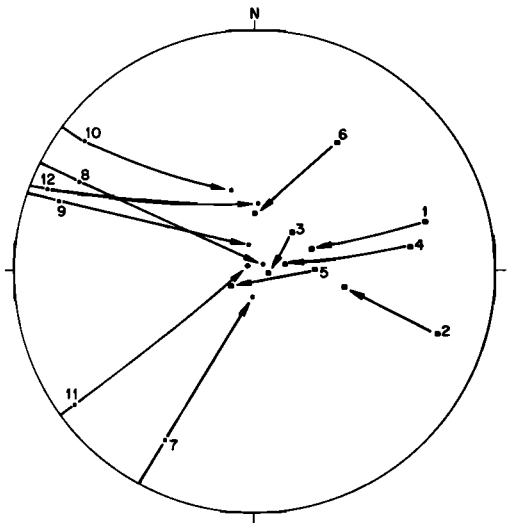



Fig. 4. Equal area projection of magnetization directions from the two limbs of the Centennial Valley syncline. Squares (numbers 1–6) refer to samples from the east limb, and circles (7–12) to samples from the west limb. Open symbols represent upper hemisphere directions. Arrows indicate change of direction resulting from correction for local folding.

existed during deposition of these rocks, (2) that the shale bedding planes were originally dipping several degrees toward the southeast, and (3) that northwesterly currents operated during deposition to produce inclination errors [*Rees*, 1961] in the remanent directions.

## DISCUSSION

As may be seen from Figures 2 and 3, samples from this study give a fairly complete coverage of the interval from late Coniacian to early Maestrichtian (about 88–69 m.y. B.P.). None of these samples were found to have reversed polarity. This finding is in agreement with the Cretaceous paleomagnetic summary presented by *Helsley and Steiner* [1969] except for the mid-Santonian reversal reported by *Sasajima and Shimada* [1966] from the Akoh formation (Japan). More recently, however, the Akoh formation has been radiometrically dated between 92–104 m.y. (S. Sasajima, personal communication, 1973). Another reversal in sedimentary rocks has been reported by *Shmeleva* [1963] from the Agaaral suite (USSR), but the age assignment of these rocks is no more precise than lower Senonian. A more recent definition of the normal polarity interval in the Cretaceous is given by *Irving and Couillard* [1973] based mainly on the compilation of paleomagnetic data by *Hicken et al.* [1972]. This study shows that reversals should be found for that part of the Cretaceous younger than about 82 m.y., but the three mixed polarity determinations that define the end of the normal interval [*Vollstadt et al.*, 1968; *Watkins and Cambray*, 1971; *Fujiwara and Nagase*, 1965] are not well dated. The mixed polarities of the well-dated Boulder batholith [*Hanna*, 1973] do, however, fall clearly within the time interval represented by the Niobrara formation.

A result of normal magnetization from the late Coniacian to the early Maestrichtian is not in accord with the geomagnetic polarity time scale extrapolated from linear marine magnetic anomalies. The Upper Cretaceous section of the reversal time scale derived by *Larson and Pitman* [1972] shows six intervals of reversed polarity between the late Coniacian and the early Maestrichtian. We feel that there are five possible explanations that could reconcile these conflicting observations, and we discuss these briefly below:

1. The linear magnetic anomalies could have been misdated. Since the magnetic anomaly reversal time scales between the present and the Upper Cretaceous depend on the assumption that sea floor spreading rates are fairly constant over long periods of time, violation of that assumption could cause large differences between extrapolated and real ages of polarity intervals. The ages of critical anomalies can be roughly checked, however, by micropaleontological dating of sediment overlying basement in areas of the ocean floor near these anomalies. This has been done often enough by the Joides program that it now appears that the Upper Cretaceous anomaly dates are not significantly in error. We refer in particular to sediment age determinations at sites 10, 20, and 163 [*Peterson et al.*, 1970; *Maxwell et al.*, 1970; *van Andel et al.*, 1973] that show that anomalies 30, 31, and 32 are probably quite accurately placed with respect to the present zonation of the Upper Cretaceous.

2. The Niobrara formation may be misdated. If none of the Niobrara formation were younger than middle Santonian, our results would be in disagreement only with a questionable Coniacian reversal on *Larson and Pitman*'s [1972] time scale. However, the microfaunal analyses of our samples are not likely to be in error by that much. In addition, since many of the same forms were used to date the Niobrara samples as the oceanic cores, a reassignment of an older age to the Niobrara would probably mean a similar shift toward older age of the Upper Cretaceous anomalies. This possibility cannot be ruled out, however, because of the difficulty in correlating paleontological age with radiometric age [*Casey*, 1964].

3. The Niobrara formation may not have been deposited during reversed intervals, or we may have missed reversals because of a too large sampling interval. This explanation is based on bad luck and seems statistically unlikely. The magnetic anomaly time scale suggests that the magnetic field was in reversed polarity for at least one third of the time between upper Coniacian and Lower Maestrichtian. If this hypothesis is true, it is highly improbable that unconformities could have occurred at all times when the field was reversed, or that from more than 300 samples not one with reversed polarity should be found. On the other hand, strong evidence for multiple reversals in the Elkhorn Mountain volcanics [*Hanna*, 1967] and the Boulder batholith [*Hanna*, 1973] with radiometric dates from 68 to 78 m.y. B.P. definitely favors this possibility or possibility 2.

4. The Niobrara formation may have been remagnetized during times of normal polarity. It is possible that parts of the Niobrara formation have been remagnetized, although stable secondary magnetization has not been clearly documented for any of the samples. The shale fold test and the determination of reliable Cretaceous paleomagnetic poles from other units indicate that much of the formation has been stable at least since the early Tertiary. If it was remagnetized, for example, during the Maestrichtian, it is difficult to explain why parts of it were not remagnetized during reversed polarity intervals.

5. Some of the Upper Cretaceous anomalies may not represent complete reversals of the field. An anomaly could be produced if the field during an interval of normal polarity experienced a prolonged intensity decrease with no change in polarity. This suggestion is speculative but worth investigating. Such behavior may seem odd in light of our experience with the nature of the field during the last 4.5 m.y. but not in comparison with the long period of normal polarity

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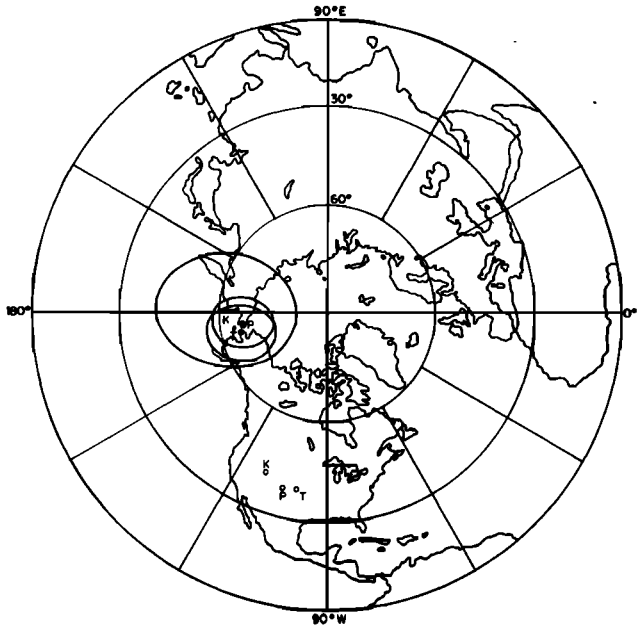



Fig. 5. Equal area northern hemisphere showing pole positions with ovals of confidence and site locations for Pueblo, Trego, and Kemmerer samples.

and apparent excursions to which it was subject during the Cretaceous.

Except for some of the chalks the completely oriented samples provide reliable data for the determination of paleomagnetic pole positions for the Upper Cretaceous. These positions are indicated in Table 1 and are plotted together with the appropriate ovals of confidence in Figure 5. These pole positions agree well with the bulk of the Cretaceous pole positions for North America [*Grommé et al.*, 1967]. They agree especially well with the pole from the Late Cretaceous Elkhorn Mountain volcanics at 69°N, 171°W [*Hanna*, 1967] and with the pole from the Sierra Nevada granites (85 m.y. B.P.) at 69°N, 165°W [*Grommé and Merrill*, 1965]. None of our samples consistently indicated anomalous low-latitude poles such as those reported from the Franciscan formation [*Grommé and Gluskoter*, 1965] and related ultramafics [*Saad*, 1969] and some Cretaceous dikes from Jamaica [*Watkins and Cambray*, 1971].

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